

HANFEI DING

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Education

Harbin Engineering University (Project 211)

Sep. 2021 – Jun. 2025

Bachelor of Engineering in Computer Science and Technology

Harbin, China

- **GPA: 90.65/100**
Achieved scores of **90+** in **30 required courses**, including **92+** in all **mathematics-related courses**.
- **TOEFL iBT: 4.5/6.0**

Relevant Coursework

- Probability and Statistics (**100**)
- Mathematical Analysis (**98**)
- Linear Algebra (**97**)
- Data Structures (**95**)
- Algorithm Design and Analysis (**93**)
- Operating Systems (**91**)
- Computer Organization (**96**)
- Artificial Intelligence (**95**)
- Discrete Mathematics (**92**)

Research Experience

Fine-Tuning SAM for Brain Tumor Segmentation

Dec. 2024

Team Leader, Undergraduate Research Assistant, Deep Intelligence Laboratory, Harbin Engineering University Harbin, China

- Adapted SAM and MedSAM to BraTS 2020 brain tumor segmentation through data-efficient and parameter-efficient fine-tuning.
- Constructed pseudo-RGB MRI inputs from FLAIR, T1ce, and T2 modalities and generated WT/TC/ET binary masks for hierarchical tumor-region segmentation.
- Fine-tuned only the Mask Decoder with stochastic box prompts and BCE-Dice loss while freezing the Image Encoder and Prompt Encoder to reduce computational cost and overfitting risk.
- Achieved an average Dice score of 0.7984 with MedSAM, outperforming SAM ViT-B and SAM ViT-H baselines, and built an interactive web-based tumor segmentation prototype.

Masked Face Recognition with Machine Learning and OpenCV

Jul. 2023

Team Leader, Visual Computing Summer Research Program, National University of Singapore

Singapore

- Developed a machine learning and OpenCV-based face recognition system for both masked and unmasked faces, addressing the performance degradation of conventional face recognition methods under facial occlusion.
- Constructed a masked-face dataset using facial landmarks extracted with dlib, and processed facial features with Gaussian blur, mean filtering, and SIFT-based feature extraction.
- Trained and compared SVM, KNN, and Random Forest classifiers, and designed a final decision model that achieved a 100% stranger interception rate, an 86% registered-user pass rate, and 85% overall recognition accuracy.

Projects

Fault Classification Tool for Microservice Systems | Python, PyTorch, GNN, Multimodal Learning

Jun. 2025

- Independently designed and implemented a multimodal fault diagnosis tool for microservice systems as an undergraduate thesis project, jointly performing anomaly detection and root cause localization from logs, traces, and metrics.
- Built a multimodal feature encoder using linear projection, dilated causal convolution, multi-head attention, and GLU-based adaptive fusion to generate service-level node embeddings.
- Constructed service invocation graphs and learned topology-aware representations with GAT and GGNN to model dynamic dependencies and long-range fault propagation.
- Developed a parameter-shared joint classifier with cascaded anomaly detection and root cause localization modules, achieving the best overall performance among multiple baselines on the Eadro dataset and receiving the **Outstanding Undergraduate Thesis (Design) Award**.

Forum Discussion and Admin Management System | Flask, MySQL, Redis, HTML, CSS, JavaScript

Nov. 2023

- Developed a Flask-based full-stack forum system supporting user registration, login, post publishing, commenting, topic-based discussion boards, and keyword-based fuzzy search.
- Implemented the frontend with HTML, CSS, and JavaScript, and built the backend with Flask, MySQL, and Redis, including an administrator system for managing users and forum content.

Awards & Honors

Competitions

- **Mathematical Modeling League of Three Northeastern Provinces** (Top 18.52%) **Jul. 2024**
Regional 1st Prize; **Team Leader / Primary Contributor**
- **Lanqiao Cup Software Competition** (Top 20%) **May 2024**
Provincial 2nd Prize
- **National English Competition for College Students, Category C** (Top 8.6%) **Oct. 2022**
National 3rd Prize
- **China Undergraduate Mathematical Contest in Modeling** (Top 0.6% among 49,000+ entries) **Sep. 2022**
Provincial 1st Prize; **Team Leader / Primary Contributor**

Scholarships & Honors

- **Outstanding Undergraduate Thesis (Design), Class of 2025** **Jun. 2025**
One of 11 recipients across the School of Computer Science and Technology, School of Software Engineering, and School of Information Security.
- **Outstanding Student Scholarship, Harbin Engineering University** **Multiple Years**
School-level Scholarship; 5-time recipient; awarded to the top 8–15% of students.
- **Merit Student, Harbin Engineering University** **2021 – 2022**
University-level Honor; awarded to the top 6% of students.

Extracurricular Activities

Underwater Robotics Club | *Harbin Engineering University* **Nov. 2021 – Sep. 2022**

- Participated in team-based robotics activities, gaining hands-on exposure to underwater robot design and engineering practice.

Freshman Debate Competition | *Best Debater* **Oct. 2021**

- Recognized for logical argumentation, public speaking, and structured communication.

Campus Volunteer Activities | *Energy Conservation and Anti-Epidemic Service* **Sep. 2021**

- Supported campus volunteer initiatives related to public-awareness promotion and community service.

Skills & Interests

Programming Languages: C, C++, Python, Java
Frameworks & Tools: PyTorch, OpenCV, NumPy, pandas, scikit-learn, Matplotlib, Flask, MySQL, Git, Docker, Linux
Interests: Swimming, badminton, fitness training, LEGO model building